by sodium hydroxide. The finish is permanent in nature and is also known as Organdi finish or Parchmentisation.

1.3.7 Softening

Softening of fabric is a very important functional finish which is required to give a pleasant hand feel to the fabric and impart better drapability. Fabrics that are harsher and stiffer because of their construction or due to some prior finishing process are softened by this method. Softening can be done by either mechanical or chemical process. Simple calendaring softens hand of the fabric, but is temporary in nature. Silicone compounds are mostly used as softeners which are durable and require curing. The different types of softeners are anionic softeners, cationic softeners, non-ionic softeners, reactive softeners, emulsion softener and silicon softeners. Other types include emulsified oils and waxes which often result in a semi-durable finish.



Figure 1.7 Fabric with Bad Drape at Left and Fabric with Good Drape at Right

Anionic softeners are not fast to wash. They are compatible with resin and used as a temporary finish with starch and cationic product.

Example: Sulphonated oils and fatty alcohol sulphates. Non-ionic softeners have excellent stability against yellowing and are not fast to dyeing. Cationic softeners are substantive to cellulosic material and therefore, remain on cloth for few washes. They are compatible with resins. Reactive softeners are more durable softeners and react chemically with the (–OH) groups of cellulose. They are more expensive and also toxic in nature. Emulsion softeners are more popular because it reduces the loss of tear strength on resin finish and are fast to washing. Silicon softeners are the most used softeners in recent times. These are the manmade polymers based on the framework of alternate silicon and oxygen bonds with organic substituent are attached to silicone.

1.3.8 Shearing

Shearing is a process that evens out the length of the pile of a fabric in a controlled manner. It removes the surface fibres, yarn ends, knots and similar irregularities and surface flaws with the help of cropping or cutting. The fabric is passed through a series of tension bars and over an angled shearing bed which uses blades to cut the protruding fibres. The shearer head consists of spiral blade which is in contact with a ledger blade. The fabric is wound helically around a rotating cylinder which moves around spiral blade and ledger blade. Strong suction is used to remove the cut fibres from the machine.

The distance to move the bed of ledger blade is adjustable and the height of the pile can be regulated. Shearing may also create a smooth surface or a patterned or sculptured effect by flattening portions of the pile with an engraved roller and shearing off the areas that remain erect, and steaming the fabric to raise the flattened or taller portions. Thus Shearing can be used to create raised patterns or to smooth the overall nap of a fabric.

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